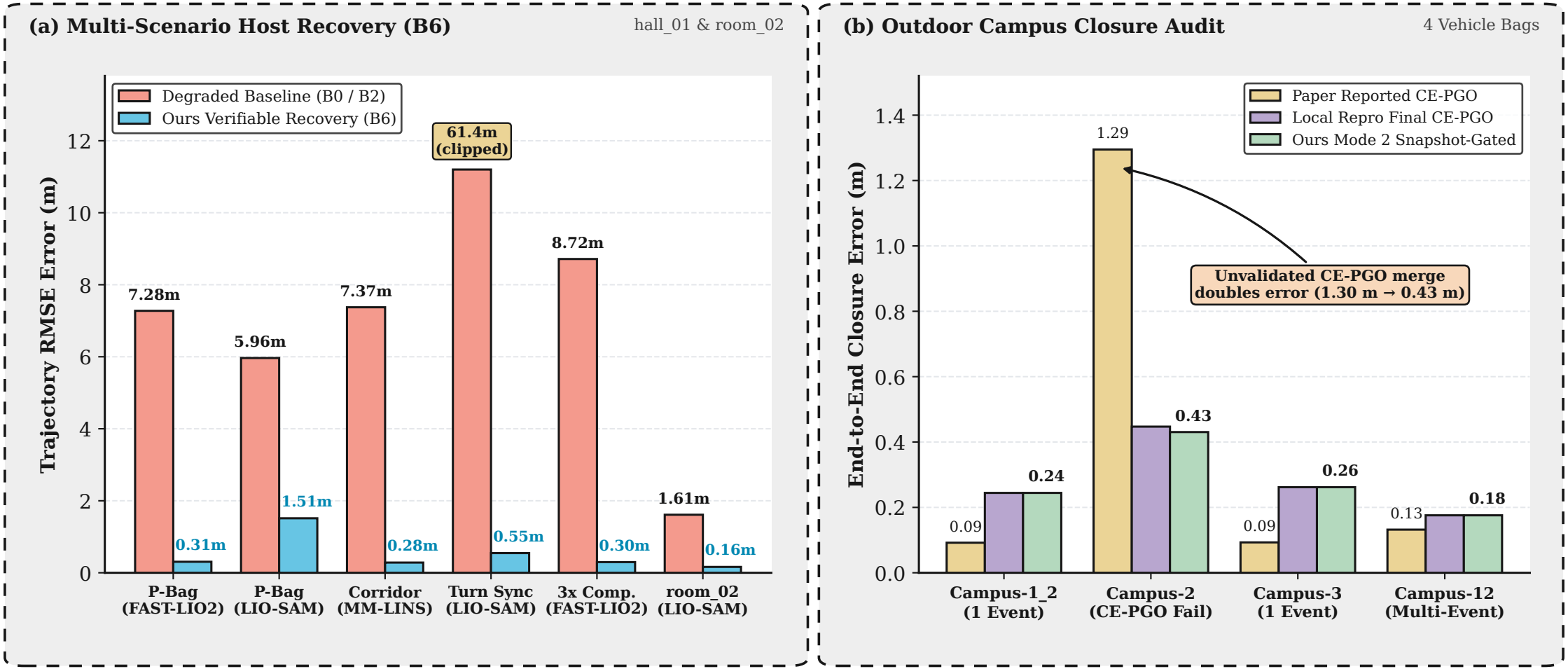
(a) Multi-Scenario Host Recovery (B6)
hall_01 & room_02
Degraded Baseline (B0 / B2)
Ours Verifiable Recovery (B6)
Trajectory RMSE Error (m)
0
2
4
6
8
10
12
7.28m
0.31m
5.96m
1.51m
7.37m
0.28m
61.4m (clipped)
0.55m
8.72m
0.30m
1.61m
0.16m
P-Bag (FAST-LIO2)
P-Bag (LIO-SAM)
Corridor (MM-LINS)
Turn Sync (LIO-SAM)
3x Comp. (FAST-LIO2)
room_02 (LIO-SAM)
(b) Outdoor Campus Closure Audit
4 Vehicle Bags
Paper Reported CE-PGO
Local Repro Final CE-PGO
Ours Mode 2 Snapshot-Gated
End-to-End Closure Error (m)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0.09
0.24
1.29
0.43
0.43
0.09
0.26
0.26
0.13
0.18
0.18
Unvalidated CE-PGO merge doubles error (1.30 m → 0.43 m)
Campus-1_2 (1 Event)
Campus-2 (CE-PGO Fail)
Campus-3 (1 Event)
Campus-12 (Multi-Event)